State Estimation and Filtering for Robotic Perception

AA273

Markovianity, Recursive Bayesian filtering,

Gaussian RVs, Mean, Covariance



Stanford University

MSL

AA273: State Estimation and Filtering for Robotic Perception

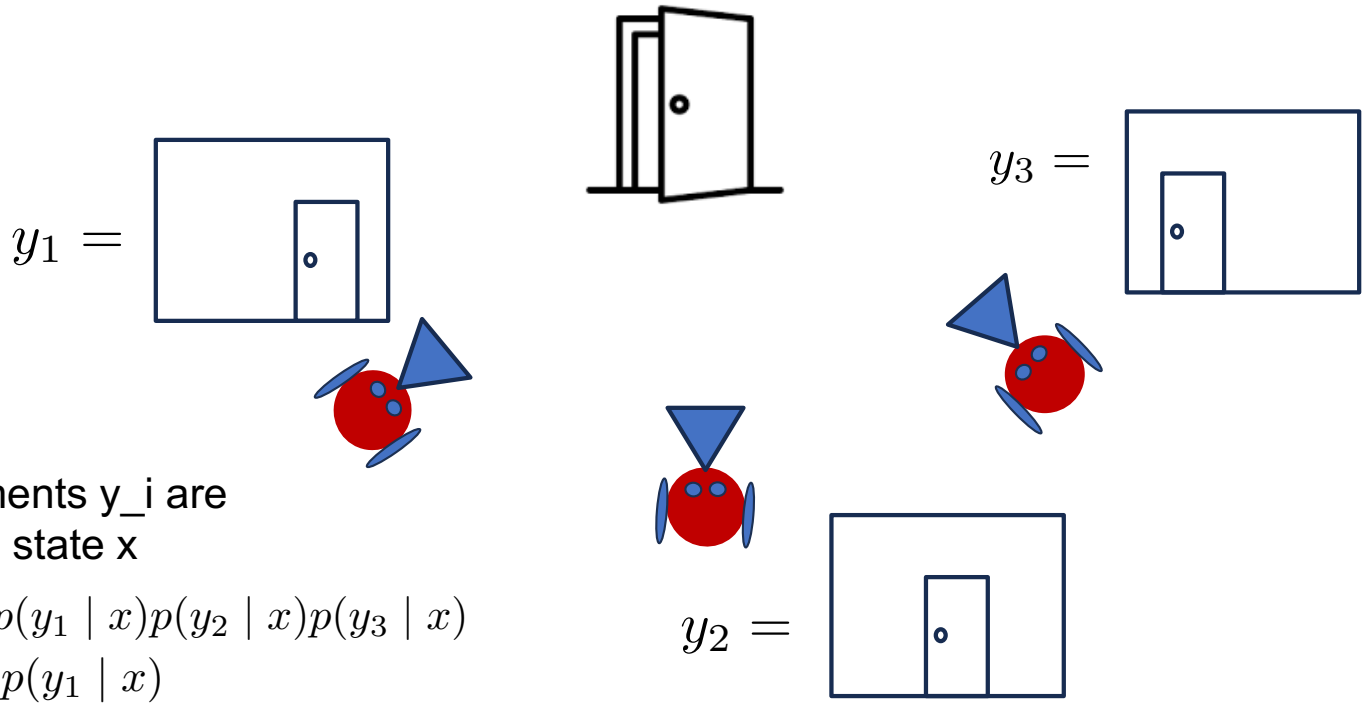
Announcements:

- HW 1 due Fri Jan 16
- HW 2 out Fri Jan 16, due Fri Jan 23
- Waitlisted students: email me if you are still on waitlist and still interested in taking the class. Waitlist currently down to 8
- Project topic ideas posted
- Will soon post group signup sheet
- New CA: Graham Shunk: OHs and contact info will be posted on Canvas in an updated Syllabus

Lecture 2:

- Markovianity
- Recursive Bayesian filter
- Multivariate Gaussian RVs, Mean, Covariance

Conditionally Independent Measurements in Robotics



Claim: Measurements y_i are cond. indep. given state x

$$p(y_1, y_2, y_3 \mid x) = p(y_1 \mid x)p(y_2 \mid x)p(y_3 \mid x)$$
$$p(y_1 \mid x, y_2, y_3) = p(y_1 \mid x)$$
$$p(y_2 \mid x, y_2, y_3) = p(y_2 \mid x)$$
$$p(y_3 \mid x, y_1, y_2) = p(y_3 \mid x)$$

Conditionally Independent Measurements

Define shorthand:

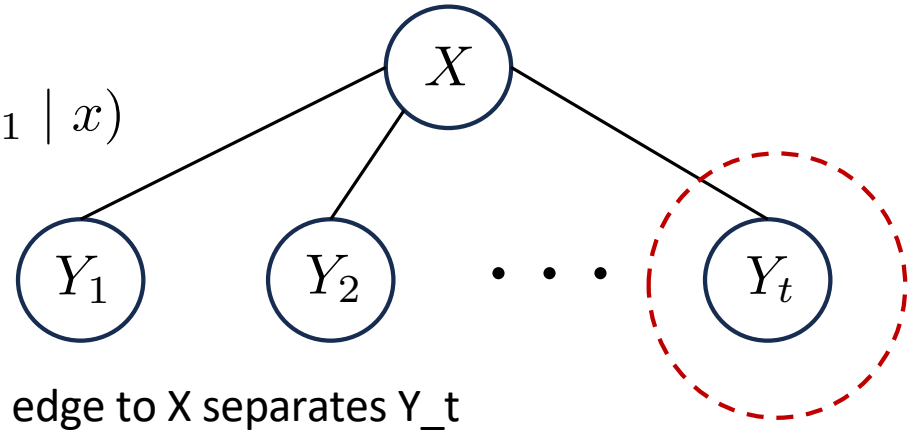
$$y_{1:t} := (y_1, y_2, \ldots, y_t)$$

Factor the likelihood:

$$p(y_{1:t} \mid x) = p(y_t \mid y_{1:t-1}, x) p(y_{1:t-1} \mid x)$$

$$= p(y_t \mid x)$$

$$p(y_{1:t} \mid x) = p(y_t \mid x) p(y_{1:t-1} \mid x)$$



Cutting edge to X separates Y_t from graph:

$$p(y_t \mid y_{1:t-1}, x) = p(y_t \mid x)$$

Simplify:

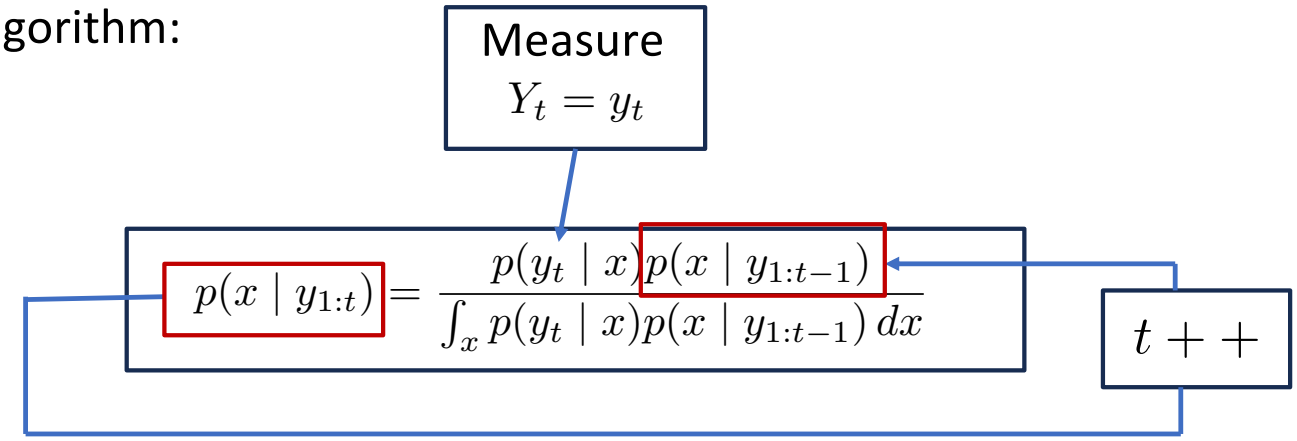
$$p(y_{1:t} \mid x) = \prod_{\tau} p(y_\tau \mid x)$$

Sequential Bayes' Rule

$$p(x \mid y_{1:t}) = \frac{p(y_t \mid x)p(x \mid y_{1:t-1})}{\int_x p(y_t \mid x)p(x \mid y_{1:t-1})\, dx}$$

(Derivation in notes / on board)

Recursive algorithm:



Random Processes

Collection of jointly distributed, time indexed RVs:

$$X_{1:t} = (X_1, X_2, \ldots, X_t)$$
$$X_{1:t} \sim p(x_{1:t})$$

Usually defined by special temporal correlation, or lack of temporal correlation.

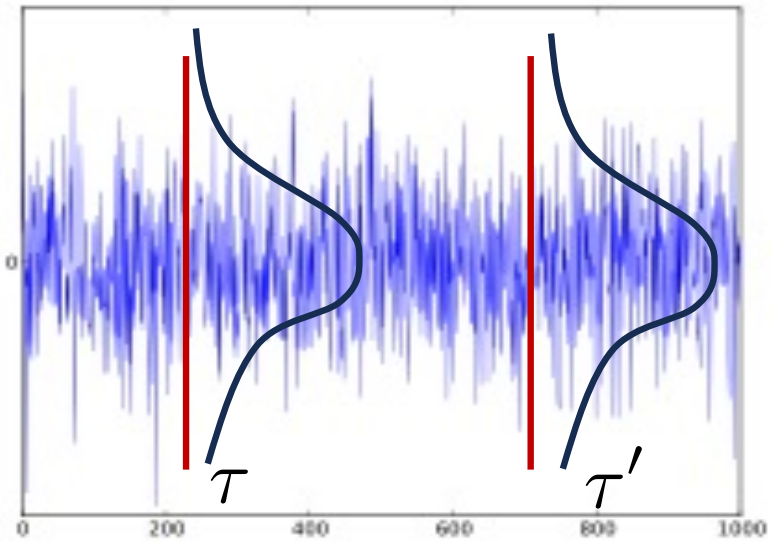
E.g., white noise process (no temporal correlation)

$$p(x_{1:t}) = \prod_{\tau} p(x_\tau) \quad \text{equivalently} \quad p(x_t \mid x_\tau) = p(x_t) \quad \forall \tau \neq t$$

E.g., stationary white noise process (a.k.a, independent and identically distributed)

$$p(x_t \mid x_\tau) = p(x_t) \quad \forall \tau \neq t \quad \text{and} \quad p(x_t) = p(x_\tau) \quad \forall\, t, \tau \quad \text{equivalently} \quad X_{1:t} \quad \text{i.i.d.}$$

E.g. Gaussian White Noise Process



Gaussian:

$$X_t \sim \mathcal{N}(\mu_t, \Sigma_t)$$

White noise:

$$p(x_t \mid x_\tau) = p(x_t) \quad \forall \tau \neq t$$

We write:

$$X_t \sim \mathrm{GWN}(\mu_t, \Sigma_t)$$

Markov Random Processes

- One step time-correlation.
- “The future is independent of the past given the present.”
- Analogous to concept of state in dynamics and control.

Def (Markov): A random process $X_{0:t}$ is Markov iff

(i) $p(x_t \mid x_{0:t-1}) = p(x_t \mid x_{t-1})$

(ii) $p(x_\tau \mid x_{0:\tau-1,\tau+1:t}) = p(x_\tau \mid x_{\tau-1}, x_{\tau+1}) \quad \text{for } 0 < \tau < t$

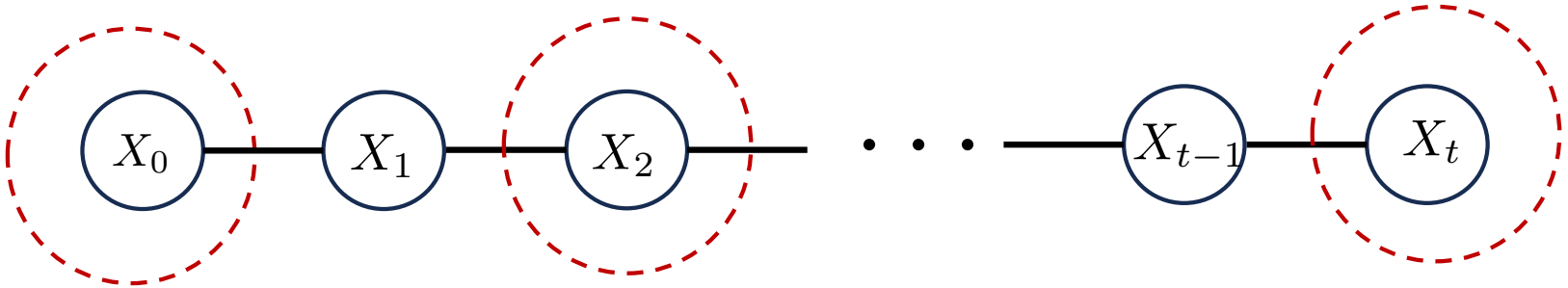
(iii) $p(x_0 \mid x_{1:t}) = p(x_0 \mid x_1)$

- Is a form of conditional independence!

Markov Process Graphical Model

$$p(x_0 \mid x_{1:t}) = p(x_0 \mid x_1)$$

$$p(x_t \mid x_{0:t-1}) = p(x_t \mid x_{t-1})$$



$$p(x_\tau \mid x_{0:\tau-1,\tau+1:t}) = p(x_\tau \mid x_{\tau-1}, x_{\tau+1})$$

Notation: $:= x_{-\tau}$

Standard Markov factorization

$$p(x_{0:t}) = p(x_t \mid x_{t-1})p(x_{0:t-1})$$

$$= p(x_t \mid x_{t-1})p(x_{t-1} \mid x_{t-2})p(x_{0:t-2})$$

$$\vdots$$

$$p(x_{0:t}) = \prod_{\tau=1}^{t} p(x_\tau \mid x_{\tau-1})p(x_0)$$

transition distribution

prior

Hidden Markov Model/State Space Model

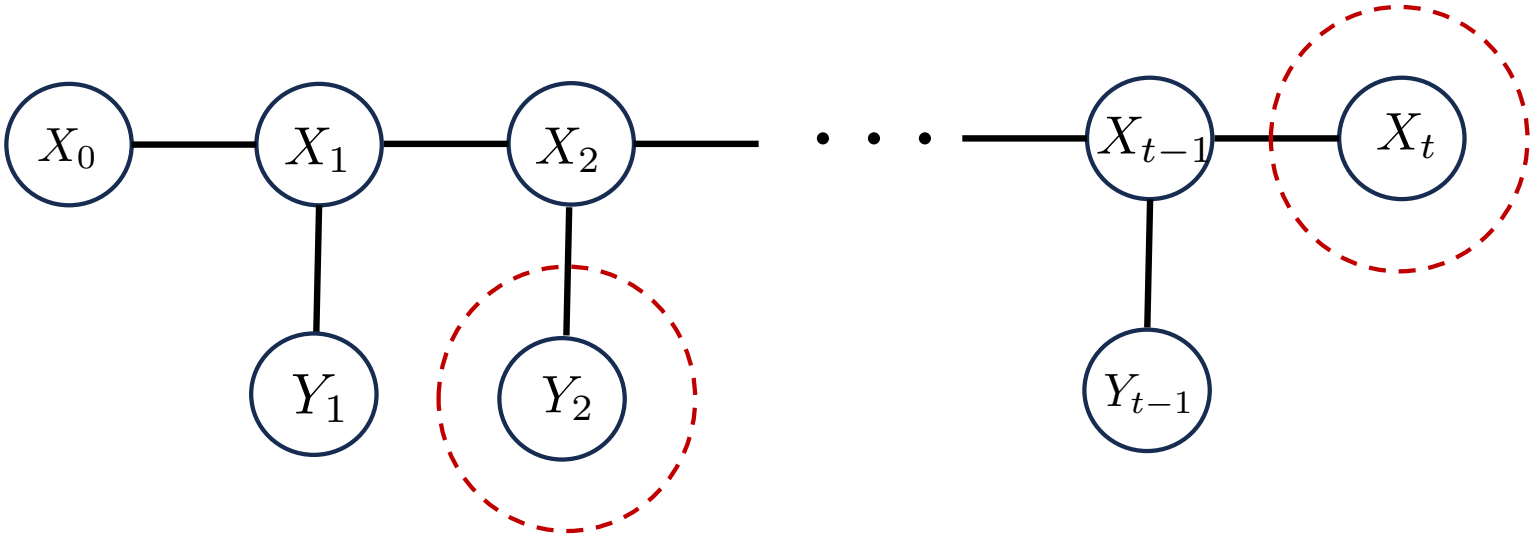
Key Assumptions:

(i) Conditionally independent measurements

(ii) Markovianity

Transition dist.

$$p(x_t \mid x_{0:t-1}, y_{1:t-1}) = p(x_t \mid x_{t-1})$$



Likelihood $p(y_\tau \mid x_{0:t}, y_{-\tau}) = p(y_\tau \mid x_\tau)$

Recursive Bayesian Filter

Goal: find posterior belief $p(x_t \mid y_{1:t})$ as a function of $p(x_{t-1} \mid y_{1:t-1})$.

prior likelihood transition dist.

Given: $p(x_0),\ p(y_t \mid x_t),\ p(x_t \mid x_{t-1})$

prediction dist

Prediction: $p(x_t \mid y_{1:t-1}) = \int_{t-1} p(x_t \mid x_{t-1}) p(x_{t-1} \mid y_{1:t-1})\, dx_{t-1}$

$t++$

Update: $p(x_t \mid y_{1:t}) = \dfrac{p(y_t \mid x_t) p(x_t \mid y_{1:t-1})}{\int_{x_t} p(y_t \mid x_t) p(x_t \mid y_{1:t-1})\, dx_t}$

posterior/belief

Derivation on the board and in the notes.

Histogram Filter / Discrete Bayesian Filter

On the board and in the notes.

Multivariate Gaussian RVs

RV takes on vector values: $x \in \mathbb{R}^n$

Def. of multi. var. Gaussian:

$$p(x) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)}$$

Shorthand: $X \sim \mathcal{N}(\mu, \Sigma)$

Parameterized by:

Mean: $\mu \in \mathbb{R}^n$

Covariance: $\Sigma \in \mathbb{R}^{n \times n}$

Symmetric, positive semi-definite

$\Sigma^T = \Sigma \qquad x^T \Sigma x \geq 0 \quad \forall x$

